

CodeAlpha Data Analytics Internship

# **Task 2 — Exploratory Data Analysis**

## **Task 3 — Data Visualization**

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Project: Ivom Retail Stores — FY2024 Sales Performance Analysis

Dataset: 971-row FY2024 POS transaction export (multi-branch, multi-category retail)

Tool used: Microsoft Excel — PivotTables, PivotCharts, dashboard

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## TASK 2 · EXPLORATORY DATA ANALYSIS (EDA)

### 2.1 Meaningful questions asked before analysis

Before opening a single PivotTable, the analysis was framed around seven business questions the (fictional) Regional Operations Director needed answered:

- 1 Which branches and states are driving the most revenue, and which are underperforming?
- 2 Which product categories and individual products sell best, by both revenue and quantity?
- 3 How does monthly sales performance trend across the year — is there seasonality?
- 4 How much revenue is being lost to discounts, and which customer type receives the most discount?
- 5 Retail vs Wholesale vs Corporate — which customer segment is most valuable to the business?
- 6 Which payment methods are customers actually using, and does this vary by branch?
- 7 Are there data quality problems in the export that could be distorting any of the figures above?

### 2.2 Data structure — variables and types

The raw export contained 971 transaction rows across 14 columns. After removing 15 exact duplicate rows, 956 unique transactions remained for analysis.

| Column                                     | Type                            | Notes                                                                 |
|--------------------------------------------|---------------------------------|-----------------------------------------------------------------------|
| Order ID                                   | Text (unique ID)                | One row per transaction                                               |
| Order Date                                 | Date                            | Spans Jan–Dec 2024                                                    |
| Branch / State                             | Categorical (text)              | 8 branches across 5 states; casing/spacing inconsistent in raw export |
| Category / Product                         | Categorical (text)              | 4 categories; casing/spacing inconsistent in raw export               |
| Quantity                                   | Numeric (integer)               | 6 rows had negative values (miscoded returns)                         |
| Unit Price / Gross Sales / Net Sales (NGN) | Numeric (currency)              | Gross = Qty x Unit Price                                              |
| Discount (%) / Discount Amount             | Numeric (percentage / currency) | 55 rows missing a Discount % value                                    |
| Customer Type                              | Categorical (text)              | Retail, Wholesale, or Corporate                                       |
| Payment Method                             | Categorical (text)              | 21 rows missing                                                       |
| Salesperson                                | Categorical (text)              | 32 rows missing                                                       |

### 2.3 Trends and patterns identified

- **Branches/states:** Benue is the top state (NGN 74.7M) simply because it runs 3 branches; by single branch, Abuja Wuse (NGN 33.7M) is actually the strongest performer. Otukpo Branch, Lagos Lekki, and especially Makurdi Main (NGN 17.5M — under half of Abuja Wuse) sit below the network's average branch revenue of NGN 27.6M and are the weakest performers.
- **Categories/products:** Electronics earns the most revenue (NGN 79.3M) from fewer units (1,403); Home & Living sells the most units (1,863). Cooking Gas Cylinder is the #1 seller by volume; Foam Mattress the #1 by revenue.
- **Seasonality:** Revenue swings from a June low (NGN 12.6M) to a December high (NGN 26.8M) — a clear holiday-driven peak.

- **Customer segments:** Retail drives the most total revenue and order volume (644 orders) but the lowest average order value (NGN 161K). Wholesale has far fewer orders (227) but the highest average order value (NGN 434K) — the most valuable segment per transaction.
- **Payment methods:** Fairly even split nationally across Bank Transfer, Cash, Mobile Money and POS/Card, though individual branches lean differently (e.g. Lagos Ikeja leans hardest to POS/Card).

## 2.4 Hypotheses tested and validated

| Hypothesis                                                                          | Result                                                                                                                                                |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Wholesale customers receive a proportionally larger discount than Retail customers. | Validated — Wholesale's discount rate (~1.5% of its gross sales) is roughly 2.5x Retail's (~0.6%).                                                    |
| Sales show holiday seasonality, peaking in December.                                | Validated — December is the highest month (NGN 26.8M); June is the lowest (NGN 12.6M).                                                                |
| The segment with the most orders is also the most valuable per order.               | Rejected — Retail has the most orders (644) but the lowest average order value; Wholesale has fewer orders (227) but the highest average order value. |

## 2.5 Data quality issues detected

- **Duplicates:** 15 exact duplicate transaction rows were found and removed (971 → 956 clean rows).
- **Miscoded returns:** 6 rows carried negative quantities — the export's way of representing returns (total -24 units, -NGN 1,393,200 in gross sales). These are netted into every total in the workbook's PivotTables, including the NGN 220,530,000 Gross Sales figure used throughout this report — they were not stripped out separately. Netting returns against gross sales is standard practice, so the totals above are valid as reported; the point worth flagging is only that they are netted in, not excluded, so a "gross sales before returns" view would read NGN 221,923,200 instead.
- **Missing values:** Discount % (55 rows), Payment Method (21 rows) and Salesperson (32 rows) all had gaps — each under 6% of rows, but enough to explain the small "(blank)" slice that shows up in a couple of PivotTables.
- **Chart-level anomaly:** Several of the dashboard's PivotCharts (Branches & State, Discount, Monthly Performance, Customer Segment) plot the PivotTable's own "Total Result" / Grand Total row as if it were just another category. Because the grand total is naturally far larger than any single branch, month, or segment, it dwarfs the real bars on the same scale — the December peak, for instance, looks almost flat next to the Total Result column. This doesn't change the underlying numbers, but it is worth fixing in a v2 by excluding the Grand Total row from each chart's source range.

## TASK 3 · DATA VISUALIZATION

### 3.1 Tool and approach

Visualization was built natively in Microsoft Excel rather than Matplotlib/Seaborn/Tableau — PivotTables were used to aggregate the cleaned 956-row dataset, PivotCharts to visualize each aggregation, and all six charts were then laid out together on a single dashboard sheet so the whole FY2024 story reads in one view.

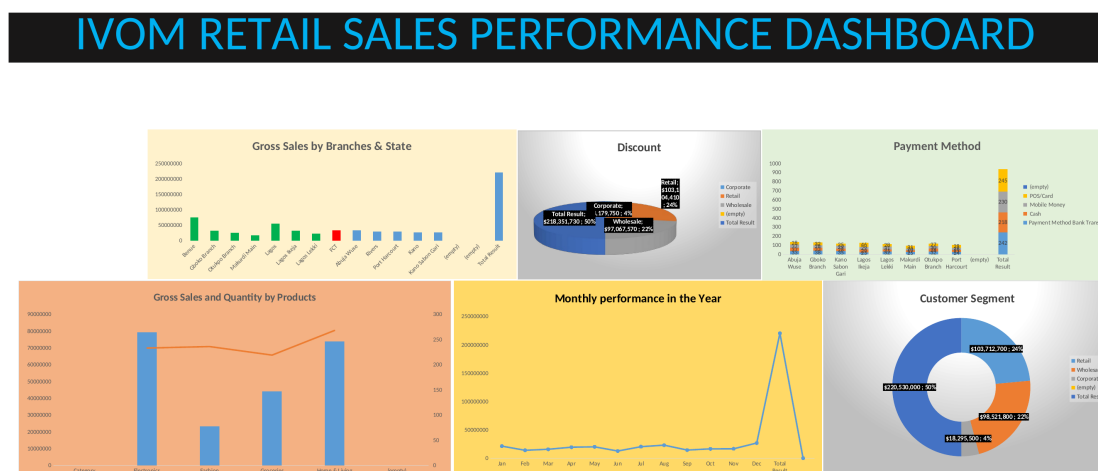
### 3.2 Dashboard walkthrough

- **Gross Sales by Branches & State** (bar) — ranks all 8 branches side by side, immediately surfacing Abuja Wuse as the top performer and Makurdi Main as the weakest.
- **Discount** (pie, by customer type) — shows Wholesale absorbing the largest single slice of total discount given.
- **Payment Method** (stacked bar, by branch) — shows the Bank Transfer / Cash / Mobile Money / POS-Card mix at each branch side by side.
- **Gross Sales and Quantity by Products** (combo bar + line, by category) — bars show revenue, the overlaid line shows units sold, making it easy to see that Electronics earns more per unit than it sells in volume, while Home & Living is the reverse.
- **Monthly Performance in the Year** (line) — traces the Jan–Dec revenue curve and the June trough / December peak.
- **Customer Segment** (donut) — shows Retail and Wholesale as the two dominant segments by revenue share, with Corporate a distant third.

### 3.3 Design choices

Each chart sits in its own colour-coded tile so the eye can separate sections at a glance, key metrics (Total Net Sales, Total Discount, Total Quantity) sit above the charts as quick-reference numbers, and chart types were matched to the question being asked — a ranked bar for branch comparison, a line for a trend over time, a donut for share-of-total, and a combo chart where two different units (revenue and units sold) needed to appear on the same axis.

### 3.4 Dashboard



Dashboard exactly as built in the workbook (native Excel PivotCharts, rendered here as a static image for the PDF). Note: the workbook's slicer control — an interactive filter button — is an Excel-only feature and does not carry over into a flattened image or PDF; the chart data itself is unchanged.